Software Test Plan (STP) - Attrition Analysis System (AAS)

Project: Attrition Analysis System - an application for organizations to track and analyse employee attrition

Version: 1.1

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Status: In progress: updated during Sprint 2

Changes in 1.1: Tools and test cases updated to match the sprints; execution status added to section 13.1. Reasons are in docs/DEVIATIONS.md.

1. Introduction

Purpose: This document defines the test plan for the Attrition Analysis System (AAS) v1.0. It outlines the objectives, scope, strategy, resources, schedule and responsibilities for testing.

Scope: Testing covers authentication and access control, organization setup, employee records (manual, CSV, HRMS sync), separation management, exit surveys, attrition analytics, flight-risk scoring, alerts and reports, plus performance, security, usability and compatibility. Client HRMS internals and the third-party email provider are excluded.

References: AAS SRS v1.0, AAS Software Architecture & Design Specification v1.0, OWASP Testing Guide v4.2, WCAG 2.1, DPDP Act 2023.

Definitions: AAS (Attrition Analysis System), HRMS (Human Resource Management System), SRS (Software Requirements Specification), RTM (Requirements Traceability Matrix), UAT (User Acceptance Testing), MFA (Multi-Factor Authentication).

2. Test Items

- Auth & RBAC module
- Organization & user management module
- Employee module (CRUD, CSV import, HRMS sync)
- Separation management module
- Exit survey module
- Analytics engine and dashboard
- Risk scoring service
- Report, alert and notification module
- REST API (all /api endpoints)

3. Features to be Tested

Features mapped to SRS requirement IDs:

- AAS-F-001: authenticate users with email and password and
- AAS-F-002: enforce role-based access (HR Admin, HR Manager, Department Manager, Leadership) and restr
- AAS-F-003: allow an HR Admin to register an organization and configure departments, locations, grades
- AAS-F-010: allow HR users to create, view, update and deactivate employee profiles (employee ID, name
- AAS-F-011: import employee records in bulk from a CSV file of up to 10,000 rows and report row-level
- AAS-F-012: synchronise employee master data nightly from a connected HRMS through its REST API
- AAS-F-013: store a monthly headcount snapshot per department to compute average headcount for any per
- AAS-F-020: allow an HR Manager to record a separation with resignation date, last working day, separa
- AAS-F-021: reject a separation whose last working day is before the resignation date, or for an emplo
- AAS-F-022: generate an exit clearance checklist (asset return, knowledge transfer, access revocation
- AAS-F-023: allow a pending resignation to be withdrawn and record it as a 'retained' case that is exc
- AAS-F-030: automatically email the exiting employee a one-time, tokenised exit-survey link when a sep
- AAS-F-031: let the employee submit the survey (1-5 ratings for manager, pay, growth, work-life balanc
- AAS-F-032: send up to 2 reminder emails at 3-day intervals if the survey is not submitted
- AAS-F-040: compute the attrition rate for a selected period as (separations in period / average headc
- AAS-F-041: show attrition broken down by department, location, tenure band (<1, 1-3, 3-5, >5 years)
- AAS-F-042: display a 12-month attrition trend chart compared with the same months of the previous yea
- AAS-F-043: rank exit reasons by frequency and show average exit-survey scores per category
- AAS-F-044: report the early-attrition rate (employees leaving within 12 months of joining)
- AAS-F-050: compute a weekly flight-risk score (0-100) for each active employee from tenure, months si
- AAS-F-051: display the top 3 factors contributing to each employee's risk score
- AAS-F-052: allow managers to log retention actions (e.g. career discussion, compensation review) agai
- AAS-F-060: alert HR Managers and the relevant Department Manager (in-app and email) when a department
- AAS-F-061: export any dashboard view as PDF and the underlying data as CSV
- AAS-F-062: email a scheduled monthly attrition summary report to subscribed users
- AAS-NF-001: Performance
- AAS-NF-002: Scalability
- AAS-NF-003: Reliability
- AAS-NF-004: Usability / Accessibility
- AAS-NF-005: Maintainability
- AAS-NF-006: Compatibility
- AAS-NF-007: Compliance / Data retention
- AAS-SR-001: TLS 1.2+ shall be mandatory for all network connections, with HSTS enabled
- AAS-SR-002: Passwords shall be stored using bcrypt (cost >= 12); an account shall be locked
- AAS-SR-003: Personal data fields (name, email, phone, salary band, gender) and survey free t
- AAS-SR-004: Every API request shall be authorised server-side by role and organization ID; a
- AAS-SR-005: The system shall keep an append-only audit log of create/update/delete actions o
- AAS-SR-006: Exit-survey responses shall be shown to Department Managers only as aggregates o
- AAS-SR-007: All inputs shall be validated and all queries parameterised (OWASP Top 10); CSV
- AAS-SR-008: Access tokens shall expire after 30 minutes of inactivity and logout shall revok

4. Features Not to be Tested

- Internal logic of client HRMS products (only our connector against a sandbox/stub is tested)
- Third-party email provider delivery infrastructure (a mail stub is used)
- Statistical validity of the risk model on real company data (needs production history; only functional behaviour is tested)
- Payroll / final settlement calculations (out of scope of the product)

5. Test Approach / Strategy

Levels:

- Unit tests - calculation and service classes (JUnit 5; Vitest for React; pytest for the risk service)
- Integration tests - API + database through MockMvc, HRMS connector against a stub, email against a mail stub
- System tests - end-to-end flows through the web UI, checked by hand in a browser
- Acceptance tests (UAT) - HR users from a pilot organization run real scenarios

Types:

- Functional testing of all features, including negative and boundary cases
- Data-accuracy testing - attrition figures compared with a manually computed spreadsheet
- Regression testing - automated suite run on every merge in CI
- Performance and load testing (JMeter)
- Security testing (see 5.1)
- Usability, accessibility (axe) and cross-browser testing

Entry Criteria: stable build deployed on the test environment, smoke test passed, seeded test data loaded, test cases reviewed.

Exit Criteria: 100 % of planned test cases executed, >= 95 % passed, 0 Critical and 0 High defects open, all High-priority requirements show status 'A' in the RTM, NFR-001..003 targets met.

5.1 Security Validation

- Verify MFA, password hashing and account lockout (SR-002)
- TLS 1.2+ and HSTS verification with testssl.sh (SR-001)
- Cross-tenant and cross-department access (IDOR) tests with two seeded organizations (SR-004)
- Encryption-at-rest check of PII columns (SR-003)
- Audit log completeness and immutability (SR-005)
- Exit-survey anonymity threshold k >= 5 (SR-006)
- OWASP ZAP active scan, fuzzing of form fields and CSV formula-injection tests (SR-007)
- Session expiry and token revocation (SR-008)
- Data retention / anonymisation job (NF-007, DPDP Act)

6. Test Environment

- **Hardware:** Cloud VM 4 vCPU / 16 GB RAM for the application, separate VM for JMeter; tester laptops with Chrome, Edge, Firefox; one tablet.
- **Software:** AAS v1.0 (Spring Boot, React, Python risk service) on H2 in PostgreSQL mode, with a PostgreSQL 16 smoke test in CI; HRMS stub and mail stub inside the test suite.

- **Tools:** JUnit 5 and MockMvc, Vitest, pytest, JaCoCo and V8 (coverage), JMeter (performance), OWASP ZAP and testssl.sh (security), axe (accessibility), Jira (defect tracking), GitHub Actions (CI).
- **Test Data:** Two synthetic organizations: 'Org A' with 50,000 employees and 5 years of history (generated with Faker) and 'Org B' with 300 employees; known separation sets for formula checks. No real personal data is used.

7. Test Schedule

Testing runs inside the two sprints (Sprint 1: 27-31 Oct 2026, Sprint 2: 02-07 Nov 2026) so that every sprint ends with a tested, working product before the presentations from 09 Nov 2026.

Milestone	Date
Test plan approved; CI pipeline with unit tests running	28-Oct-2026
Environment & synthetic test data ready	29-Oct-2026
Test case design complete (all TC-IDs in section 13.1)	30-Oct-2026
Sprint 1 testing (auth, employees, separations, dashboard KPIs)	29-Oct-2026 to 31-Oct-2026
Sprint 1 review & demo video	31-Oct-2026
Sprint 2 functional testing (survey, analytics, risk, alerts, reports)	04-Nov-2026 to 06-Nov-2026
Performance & security testing	05-Nov-2026 to 06-Nov-2026
Regression run and UAT	07-Nov-2026
Test summary report & RTM update	08-Nov-2026

8. Test Deliverables

- Test Plan (this document)
- Test cases (manual & automated) - section 13.1
- Automated test suites and JMeter plans
- Test data seeding
- Test execution logs
- Defect reports (Jira)
- Updated RTM
- Reports in docs/reports: unit, system validation, code coverage and branch coverage

9. Roles and Responsibilities

Role	Name	Responsibility
Scrum Master / QA Lead	H K Manaswin (PES1UG24AM108)	Maintain the Jira board and sprints (move issues To Do -> In Progress -> In Review -> Done), track the burndown chart, run daily check-ins, prepare this plan, coordinate execution and report status
Product Owner (HR SME)	H K Manaswin (PES1UG24AM108)	Prioritise the backlog, accept completed stories, approve test results, run UAT and sign off
Test Engineer	R Harshith (PES1UG24AM117)	Design & execute test cases, automate regression, log defects
Developer	Ganesh Aroda C (PES1UG24AM102)	Unit tests, CI pipeline, defect fixes and triage
Security & Performance Tester	Harsha R (PES1UG24AM115)	Run security validation (section 5.1) and JMeter performance tests

10. Risks and Mitigation

Risk	Mitigation
Delay in stable build delivery	Request early smoke builds; test modules as they are completed
Test environment downtime	The application and tests run on any laptop with H2, with no setup
Unrealistic synthetic data hides analytics defects	Hand-crafted datasets with known expected results
HRMS sandbox not available	Use a stub built from the vendor API specification
Limited time for performance/security testing	Schedule them in a dedicated window; automate in CI

11. Assumptions & Dependencies

- Requirements in SRS v1.0 are frozen before test case design
- Synthetic test data is accepted as representative for system testing
- HR subject-matter experts are available for UAT
- Mail and HRMS stubs behave like the real services' documented APIs

12. Suspension & Resumption Criteria

Suspend testing if:

- Test environment unavailable for more than 4 hours
- Build too unstable (blocks more than 30 % of test cases)
- A Critical security defect (e.g. cross-tenant data leak) is found - security and data tests pause until fixed

Resume testing if:

- Blocking defects are resolved and a new build passes the smoke test
- Environment is stable again
- Affected test cases are re-executed from the start

13. Test Case Management & Traceability

The RTM (SRS section 8) maps each SRS requirement to its test cases. Example:

- AAS-F-001 (Login with MFA) -> TC-AUTH-01
- AAS-F-020 (Record separation) -> TC-SEP-01, TC-SEP-02
- AAS-F-040 (Attrition rate) -> TC-ANA-01
- AAS-NF-001 (Dashboard response time) -> TC-PERF-01
- AAS-SR-004 (Tenant isolation) -> TC-SEC-04

13.1 Test Cases

Status as of 04-11-2026. Pass: an automated test covers the whole case and passed. Partial: only part is covered. Not run: not executed yet. Details are in docs/reports/system-validation-report.md.

TC ID	Req ID	Title	Steps / Input	Expected result	Type	Status
TC-AUTH-01	AAS-F-001	Login with MFA	Log in as HR Manager with valid password; enter valid OTP, then repeat with wrong OTP.	Valid OTP -> dashboard; wrong OTP -> 'Invalid code', no session.	Functional	Pass
TC-AUTH-02	AAS-F-002	Department scoping	As Sales Dept Manager call GET /analytics/attrition?dept=Finance.	HTTP 403; Finance data not returned.	Functional / Security	Pass
TC-ORG-01	AAS-F-003	Configure organization	HR Admin adds dept 'R&D', reason 'Relocation', threshold 4 %.	Values saved and appear in forms; threshold used for alerts.	Functional	Pass
TC-EMP-01	AAS-F-010	Employee CRUD	Create employee E1001; edit	Create/update succeed;	Functional	Pass

			designation; create duplicate E1001.	duplicate rejected with error.		
TC-EMP-02	AAS-F-011	CSV bulk import	Upload CSV with 1,000 rows of which 3 have invalid dates.	997 rows imported; error report lists the 3 rows.	Functional	Pass
TC-EMP-03	AAS-F-012	HRMS nightly sync	Change a record in the HRMS sandbox stub; run sync job.	Change reflected; sync log entry 'success'.	Integration	Pass
TC-EMP-04	AAS-F-013	Headcount snapshot	Run month-start job on seeded data.	Snapshot rows created per department with correct counts.	Functional	Pass
TC-SEP-01	AAS-F-020	Record resignation	Record voluntary resignation for E1001, reason 'Better pay', regretted.	201 Created; status 'Notice period'; audit entry written.	Functional	Pass
TC-SEP-02	AAS-F-021	Invalid separation dates	Last working day 01-10-2026 before resignation date 15-10-2026.	400 with field error; no record created.	Negative	Pass
TC-SEP-03	AAS-F-022	Clearance checklist	Tick 4 of 5 clearance items and close; tick the last item and close.	First close refused naming the pending item; second succeeds. Closing also needs the last working day to have passed.	Functional	Pass
TC-SEP-04	AAS-F-023	Withdraw resignation	Withdraw a resignation during the notice period.	Shown as Retained; employee active; attrition unchanged.	Functional	Pass
TC-SRV-01	AAS-F-030	Survey email	Record separation; run the send job; inspect the recording mail gateway.	One email with tokenised link within 5 min.	Integration	Pass
TC-SRV-02	AAS-F-031	Submit survey once	Submit survey via link; reuse link; use 31-day-old token.	First accepted; reuse and expired token rejected.	Functional / Negative	Pass
TC-SRV-03	AAS-F-032	Survey reminders	Leave survey unsubmitted; advance clock 9 days.	Exactly 2 reminders sent.	Functional	Pass
TC-ANA-01	AAS-F-040	Attrition rate formula	Seed 12 separations, opening HC 236, closing HC 244 for Q3.	Rate shown = 5.00 %.	Functional	Pass
TC-ANA-02	AAS-F-041	Breakdowns	View breakdown by department and tenure band for 2026.	Sum of each breakdown equals overall separations.	Functional	Pass
TC-ANA-03	AAS-F-042	Trend chart	Open 12-month trend.	12 monthly points per year; values match monthly rates.	Functional	Pass
TC-ANA-04	AAS-F-043	Exit reasons ranking	Seed reasons: pay 6, growth 4, manager 2.	Ranked pay > growth > manager with 50/33/17 %.	Functional	Pass
TC-ANA-05	AAS-F-044	Early attrition	Seed 3 leavers with tenure < 12 months out of 10.	Early-attrition share shown correctly.	Functional	Pass
TC-RSK-01	AAS-F-050	Risk score & band	Run weekly scoring job on seeded employees.	Every active employee scored 0-100; bands match thresholds.	Functional	Pass
TC-RSK-02	AAS-F-051	Risk factors	Open risk card of a High-risk employee.	Top 3 factors with contribution shown.	Functional	Pass
TC-RSK-03	AAS-F-052	Retention action	Log 'career discussion' due tomorrow; advance clock 2 days.	Action saved; shown as overdue.	Functional	Pass
TC-REP-01	AAS-F-060	Threshold alert	Set threshold 4 %; seed Sales 3-month rate 6 %.	One alert in-app + email to HR and Sales manager.	Functional	Pass
TC-REP-02	AAS-F-061	Export PDF/CSV	Export filtered dashboard.	PDF and CSV figures equal on-screen values.	Functional	Partial
TC-REP-03	AAS-F-062	Scheduled report	Subscribe user; advance clock to 2nd of month.	Summary email delivered.	Functional	Pass
TC-PERF-01	AAS-NF-001	Dashboard response time	Load script (docs/perf/load_test.py): 50k employees, 5 years of data, 50 users.	p95 dashboard <= 3 s; API p95 <= 500 ms.	Performance	Fail
TC-PERF-02	AAS-NF-002	Concurrency	Load script: 200 concurrent users on one organization.	p95 degradation < 20 %; error rate < 1 %.	Load	Partial
TC-OPS-01	AAS-NF-003	Backup & restore	Restore last nightly backup to staging.	Restore < 4 h; data loss <= 24 h.	Reliability	Not run
TC-UX-01	AAS-NF-004	Accessibility & task time	axe scan of all pages; 5 users record a separation.	No WCAG AA violations; >= 4 users finish < 3 min.	Usability	Not run
TC-UX-02	AAS-NF-006	Cross-browser	Run smoke suite on Chrome, Edge, Firefox (2 versions each).	All pass.	Compatibility	Not run
TC-QA-01	AAS-NF-005	Coverage gate	Run the test suite with JaCoCo.	Service-layer coverage >= 70 %.	Static / CI	Pass
TC-SEC-01	AAS-SR-001	TLS configuration	Check the HSTS header; run testssl.sh once a deployed environment exists.	Only TLS 1.2/1.3; HSTS header present.	Security	Partial
TC-SEC-02	AAS-SR-002	Password storage & lockout	Inspect users table; 6 wrong logins.	bcrypt hashes; account locked 15 min.	Security	Pass
TC-SEC-03	AAS-SR-003	Encryption at rest	Query employee table directly in DB.	PII columns are ciphertext.	Security	Partial
TC-SEC-04	AAS-SR-004	Tenant isolation (IDOR)	Org A token requests Org B employee/separation IDs.	404 for every request.	Security	Pass
TC-SEC-05	AAS-SR-005	Audit trail	Record a separation; read the audit log table.	Audit entry created; no API can change it (none exposes it).	Security	Pass
TC-SEC-06	AAS-NF-007	Data retention	Run anonymisation job with records	Eligible records anonymised;	Compliance	Not run

			older than 7 years.	aggregates unchanged.		
TC-SEC-07	AAS-SR-006	Survey anonymity threshold	Manager views survey results for a team with 3 responses.	'Insufficient data' shown; no individual answers.	Security	Pass
TC-SEC-08	AAS-SR-007	Injection & XSS	OWASP ZAP active scan; import CSV with '=HYPERLINK(...)'.	No High findings; cell escaped.	Security	Partial
TC-SEC-09	AAS-SR-008	Session expiry	Idle 31 min; reuse token after logout.	Both receive 401.	Security	Partial

14. Test Metrics & Reporting

Metrics collected:

- % test cases executed
- % passed / failed / blocked
- Defect density (defects per module / KLOC)
- Defect ageing and severity distribution
- Requirement coverage (requirements with >= 1 passed test)
- Automated test coverage (line and branch, from JaCoCo)

Reports:

- Daily execution status (Jira dashboard)
- Weekly defect trend
- Final Test Summary Report

15. Approvals

Role	Name	Signature / Date
QA Lead		
Dev Lead		
Product Owner		